From preferences to rewards: A brief introduction

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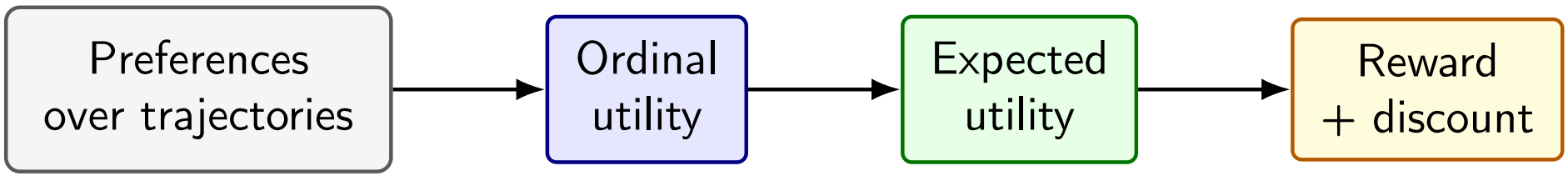
Iliad Intensive

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The Question

- Agents are powerful — but increasing capabilities lead to increasing risks.
- To understand agentic risk, we need to understand how agents work.
- Core of agency: *doing things (planning) for reasons (beliefs)*.
- What are goals? Hard question.
- **Goal of today**: understand what are preferences.

The Route



- Each arrow adds structure.
- Each structure buys a more specific representation.
- *Outcome 1*: Understand the assumptions of each step.
- *Outcome 2*: Recognise the difference between preference, utility, and reward.

Basic setting

- Let $\mathcal{O}$ be a finite observation set, and $\mathcal{A}$ a finite action set.
- A one-step interaction is

$$t = (o, a) \in \mathcal{O} \times \mathcal{A}.$$

- A finite trajectory of length n is

$$h = (o_1, a_1, o_2, a_2, \ldots, o_n, a_n) \in (\mathcal{O} \times \mathcal{A})^n.$$

- The full finite trajectory space is

$$\mathcal{H}^* = \bigcup_{n=0}^{\infty} (\mathcal{O} \times \mathcal{A})^n.$$

Preferences

- A weak preference relation $\succcurlyeq$ compares complete trajectories.

$$h \succcurlyeq h'$$

  means "h is at least as good as h'."
- From this we define:

$$h \sim h' \iff h \succcurlyeq h' \text{ and } h' \succcurlyeq h,$$
$$h \succ h' \iff h \succcurlyeq h' \text{ and not } h' \succcurlyeq h.$$

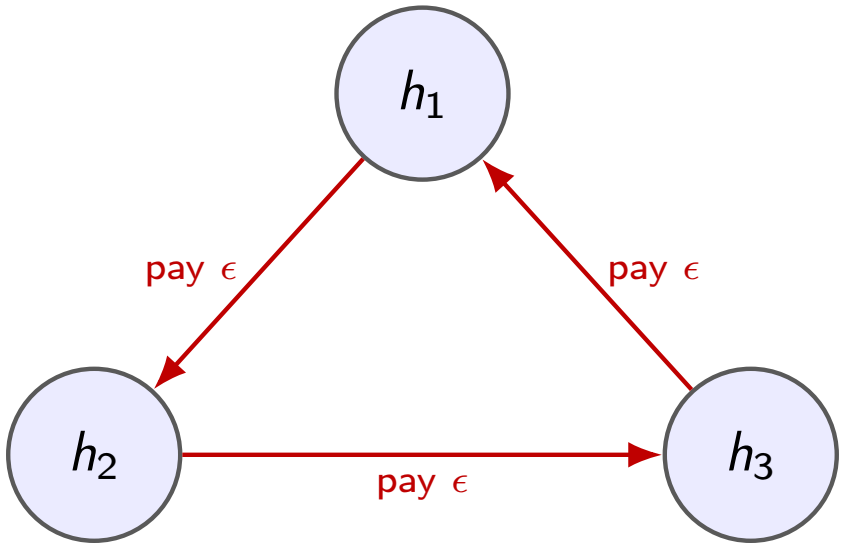
- At this point, $\succcurlyeq$ need not be complete, transitive, numerical, or 'reward-like'.

Three Kinds of Trouble

By choosing the "wrong type of preference", an agent can run into three kinds of trouble (from less to more harmful):

- **Representability failure:** preference has no convenient formal representation.
- **Self-defeat:** agents take suboptimal choices or plans — there are better alternatives that are not being taken.
  - E.g. **dominance**: another policy yields preferred trajectories at least as preferred in every state of the world, and strictly so in at least one.
- **Vulnerability:** individually acceptable trades can combine into guaranteed loss.
  - E.g. **Money pumps**: circular preferences ($h_1 \succ h_2 \succ h_3 \succ h_1$) lead to indefinite loss.

Money Pump Picture



- If the agent pays for every strict improvement, a cycle can be exploited.
- It returns to the same trajectory with less money.
- Parallel with thermodynamics: the second law of thermodynamics can be read as stating that *'nature cannot be money-pumped'*.

Three Kinds of Results

There are three types of results regarding preferences:

- **Representation results:** axioms imply that preferences can be written in a particular mathematical form.
  - E.g. some preferences can be represented as if maximising a utility function.
- **Coherence results:** violations imply some penalty for the agent.
  - E.g. money pumps or Dutch books.
- **Selection results:** some optimization pressure tends to remove agents with a given failure mode.
  - E.g. market competition or natural evolution.

These results are related (one can lead to the other) but not identical: form, vulnerability, and survival are different claims.

Axiom 1: Completeness

Completeness says every pair can be compared:

$$\forall h, h' \in \mathcal{H}^*, \qquad h \succcurlyeq h' \quad \text{or} \quad h' \succcurlyeq h.$$

- Incomparability is not the same as indifference.
- If it fails, some trajectories are genuinely incomparable.

Axiom 2: Transitivity

Transitivity says comparisons fit together:

$$h \succcurlyeq h' \text{ and } h' \succcurlyeq h'' \quad \Longrightarrow \quad h \succcurlyeq h''.$$

- If it fails, local pairwise judgments need not form a global ranking.
- The classic failure is a cycle: money pumps.

$$h_1 \succ h_2, \qquad h_2 \succ h_3, \qquad h_3 \succ h_1.$$

- Similar to Helmholtz-Hodge decomposition: transitivity is equivalent to requiring no rotational component.

Axioms 1 + 2 $\to$ Ordinal Utility

- Completeness + transitivity make $\succcurlyeq$ a weak order (total order with equivalent objects).
- On a countable space $\mathcal{H}^*$, weak orders admit an ordinal utility: there exists a function $u : \mathcal{H}^* \to \mathbb{R}$ such that

$$h \succcurlyeq h' \quad \Longleftrightarrow \quad u(h) \geq u(h').$$

- Only the ranking matters:

$$u, \quad 2u + 17, \quad \exp(u)$$

all represent the same deterministic preferences.

Beyond Ordinal Utility

Ordinal utility ranks outcomes, but gives a structure to mixtures of them.

- Example:

$$x \succ y \succ z.$$

  How should we rank y with respect to $\frac{1}{2}x + \frac{1}{2}z$? Should they be equivalent?

Ordinal rank doesn't say how to rank mixtures, which can arise naturally out of uncertainty.

Lotteries Over Trajectories

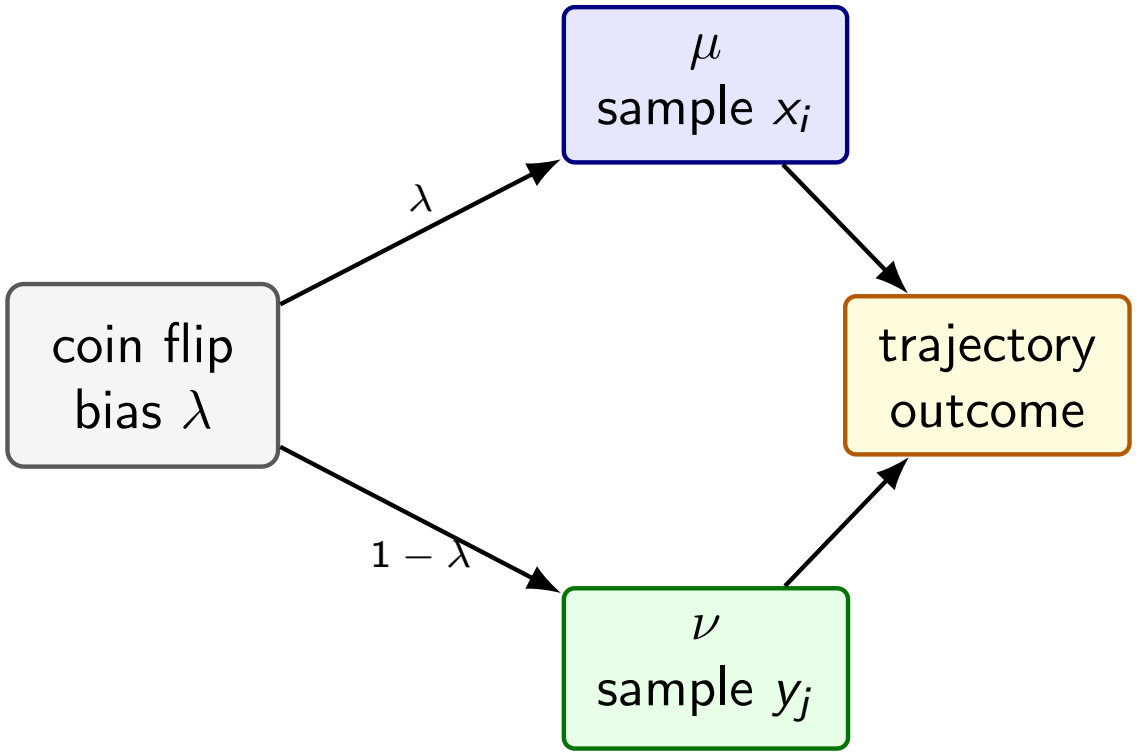
- Let $\Delta(X)$ be the set of probability distributions over $X \subseteq \mathcal{H}^*$.
- A lottery is written

$$\mu = \sum_i p_i \delta_{x_i}.$$

- A sure trajectory x is the degenerate lottery δ_x.
- We can mix lotteries:

$$\lambda\mu + (1-\lambda)\nu.$$

Lottery Picture



- vNM asks when preferences over these objects are expectations.

vNM representation: The big picture

1. **Axiom 1: Completeness:** any two lotteries can be compared.
2. **Axiom 2: Transitivity:** comparisons over lotteries fit together.
3. **Axiom 3: Continuity:** intermediate prospects can be matched by a mixture of better and worse prospects.
4. **Axiom 4: Independence:** common lottery components cancel.

Result: The first two axioms give ranking structure based on an ordinal utility; the last two axioms make the utility an expected value.

Axiom 3: Continuity

- If $\mu \succcurlyeq \nu \succcurlyeq \xi$, continuity says there always exists some value $\lambda \in [0, 1]$ such that
$$\nu \sim \lambda\mu + (1 - \lambda)\xi.$$

- Intuition: no outcome is infinitely better or worse than the others.

Classic counter-examples are lexicographic preferences, in which even the slimmest chance of a catastrophically bad outcome is worse than a slightly bad outcome with certianty, no matter how low the odds are.

Example: μ is eating chocolate, ν is eating potatoes, and ξ is accepting genocide.

Axiom 4: Independence

- If $\mu \succcurlyeq \nu$, then mixing both with the same background lottery ξ should not reverse the comparison:

$$\mu \succcurlyeq \nu \iff \lambda\mu + (1-\lambda)\xi \succcurlyeq \lambda\nu + (1-\lambda)\xi.$$

- Independence is what gives linear averaging.

When it fails, common consequences do not psychologically or behaviorally cancel.

Example (Allais paradox): Most people would say

- Getting \$100 for sure *is better than* a coin flip for \$250.
- A 1-in-100 shot at \$100 *is worse than* a 1-in-200 shot at \$250.

These are the same choice, just scaled down.

Theorem 1: The von Neumann–Morgenstern representation

Assumptions: a finite set of options $X \subseteq \mathcal{H}^*$, the four axioms hold.

- **Result 1 (existence):** there exists $u : X \to \mathbb{R}$ such that

$$\mu \succcurlyeq \nu \iff \sum_{x \in X} \mu(x)u(x) \geq \sum_{x \in X} \nu(x)u(x).$$

- **Result 2 (uniqueness):** u is unique up to positive affine transformations:

$$u'(x) = a + bu(x), \qquad b > 0.$$

Ordinal vs expected utility

Ordinal utility

- Any set of outcomes
- Preserves ranking
- Indiferent to non-decreasing transformation
- Cannot display 'meaningful gaps'

vNM utility

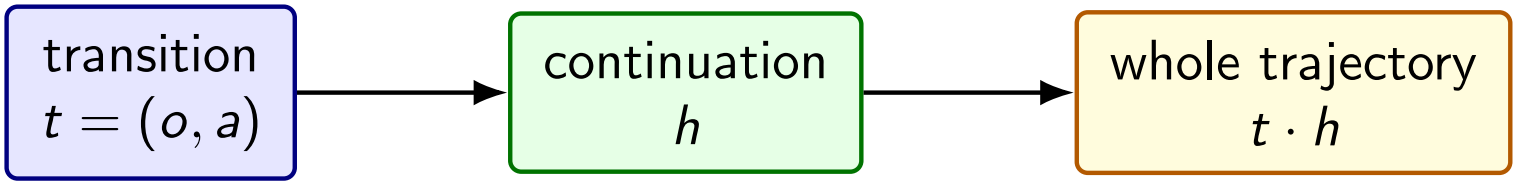
- Lotteries (probabilistic mixtures)
- Preserves risky choices
- Only affine transforms are okay
- Incommensurable gaps can exist

vNM utility further refines ordinal utility (it is not just a relabeling).

Beyond expected utility: Temporal structure

- vNM utility assigns a single scalar to a whole trajectory.
- But agents make decisions over time, so they need indications of utility that unfold step-by-step.
- From a global utility $u(h)$, one can always define the increment

$$r(t; h) = u(t \cdot h) - u(h), \quad \text{satisfying} \quad u(t \cdot h) = r(t; h) + u(h).$$



- But the increment $r(t; h)$ may be non-local, and hence cannot be optimised for directly.

Axiom 5: Temporal equanimity

Let $T = \mathcal{O} \times \mathcal{A}$ be the transition set.

- The fifth axiom says there is a function $\gamma : T \to [0, 1]$ such that prepending t rescales continuation differences, so that

$$u(t \cdot \mu) - u(t \cdot \nu) = \gamma(t)\big(u(\mu) - u(\nu)\big).$$

$\gamma(t)$ measures how much the future still matters after t.

- $\gamma(t) = 1$: future differences matter as much as present differences.
- $0 < \gamma(t) < 1$: future differences are damped.
- $\gamma(t) = 0$: after t, continuation does not matter.

This is transition-dependent discounting, not necessarily a single global constant.

Theorem 2: Markov reward representation

Assumptions: Axioms 1-5 over lotteries over trajectories.

- Then, utility has a local recursive form:

$$u(t \cdot h) = r(t) + \gamma(t)u(h), \qquad u(\varepsilon) = 0.$$

- Preferences over lotteries are still represented by expected utility.

This establishes a bridge from expected utility maximisation over whole-trajectory utility to RL-style expected sum of future reward maximisation.

Unrolling the recursion

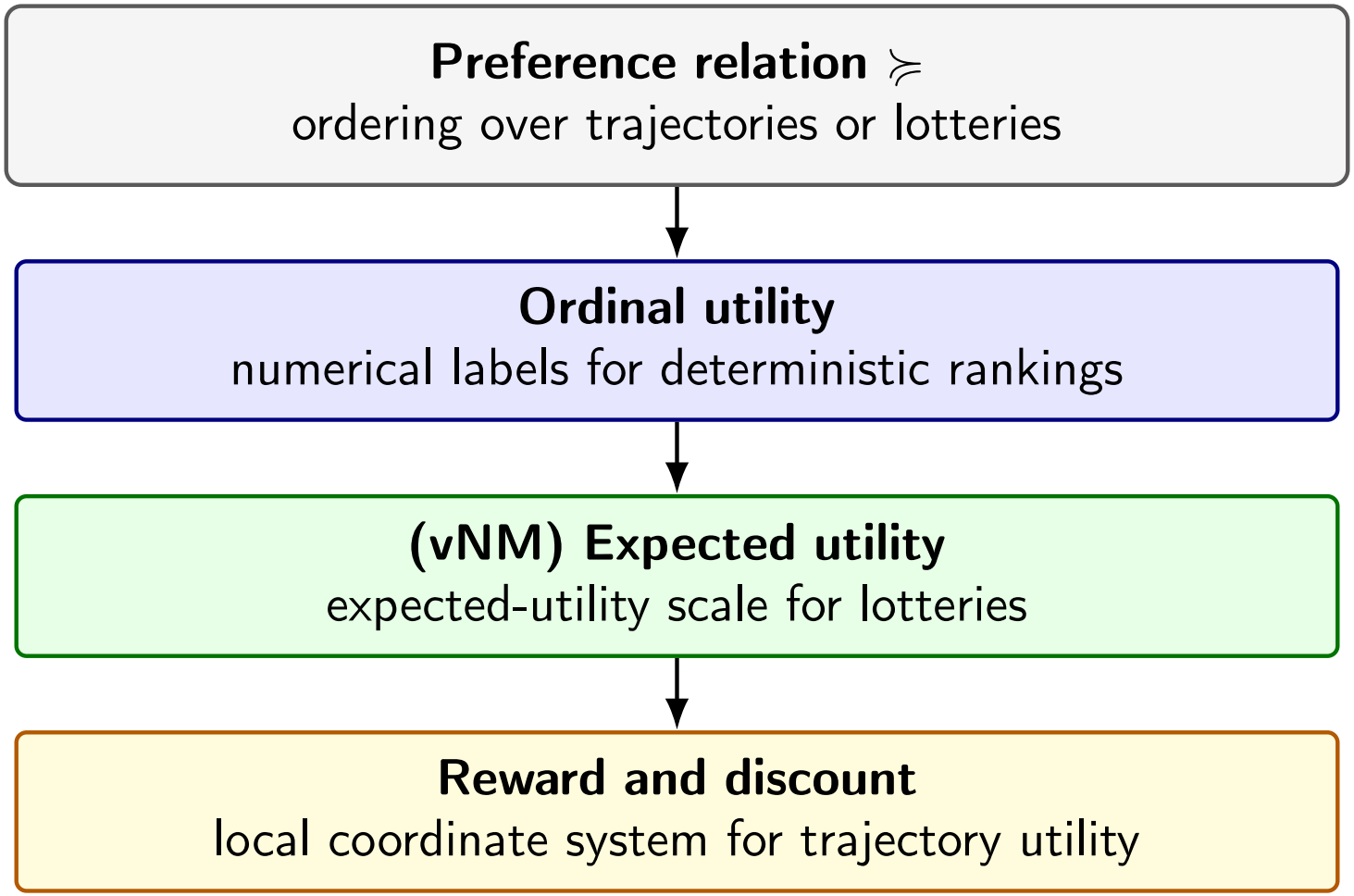
- For $h = (t_1, t_2, \ldots, t_n)$:

$$\begin{aligned} u(h) = r(t_1) &+ \gamma(t_1) r(t_2) \\ &+ \gamma(t_1)\gamma(t_2) r(t_3) + \cdots \\ &+ \left(\prod_{i=1}^{n-1} \gamma(t_i) \right) r(t_n). \end{aligned}$$

- If $\gamma(t) \equiv \gamma$, this becomes standard discounted return:

$$u(h) = \sum_{k=1}^{n} \gamma^{k-1} r(t_k).$$

Summary: Preference, utility, and reward



Open questions regarding AI alignment

- Will powerful AIs necessarily become expected utility maximisers?
- Instead of 'given a reward, how to optimise it' we could ask 'what kind of properties we want our reward to satisfy'. For example, are any of these properties (or their negation) useful for fostering:
  - Corrigibility? (e.g. incompleteness)
  - Robust values? (e.g. continuity)
  - Something else?
- Is RLHF providing an ordinal or cardinal utility signal? Are there implications of this?